

Tutor conversations: issues found and fixed

Captured (definition-capture) · 7 October 2026 · Branch `tutor-conversations` (not yet released) · Spec `Docs/tutor-conversations.md`

Issues found after the feature was built and reviewed: one while preparing the local demo, and four from the owner's first look at the new Tutor page on localhost (against the local database copy, with a demo account). Each was traced to its cause before it was changed, and each fix was checked in a browser and by the unit tests (832 passing, type check and lint clean).

#	Issue	Found by	Cause	Fix	Checked, commit
1	Search by meaning and the tutor's memory did not work: every embedding came back empty, so search silently fell back to keywords only.	Setting up the demo (seeding conversations)	The OpenRouter account's guardrails and data policy refuse the planned embedding model <code>baai/bge-m3</code> (HTTP 404, "0 endpoints available"). The app's fallback hid the failure, as designed, so nothing showed an error.	Seven multilingual models were tested against the account. The owner chose <code>qwen/qwen3-embedding-8b</code> , asked for 1024 dimensions, so the database column and its storage stay as designed. Model name and the <code>dimensions</code> parameter changed in <code>tutor.ts</code> and <code>tutorServer.ts</code> ; docs updated.	All demo answers now have vectors; a search by meaning finds related conversations. Commit <code>ad96c28</code> .
2	The sidebar had a New conversation button, which the owner did not want.	Owner, first look	It was part of the agreed design (sidebar item 2 in the spec). The owner decided conversations start only from the question box under Tutor.	Button removed from <code>TutorSidebar.tsx</code> . A new conversation starts from the empty question box on <code>/tutor</code> , which the Tutor tab opens. Sidebar test now asserts the button is absent; spec and README updated.	Browser: no button in the sidebar. Commit <code>30f4bc5</code> .
3	Searching "where does the verb go" also listed "Dative after prepositions", which says nothing about where the verb goes.	Owner, first look	The search keeps meaning matches closer than a cosine distance of 0.5. With this model, different grammar answers sit close together; the two-way prepositions answer measured 0.47 from that query, just inside the cut-off. Qwen3 expects a search query to carry a one-line task description, and the app sent the bare words.	Search queries (sidebar and the tutor's memory) now carry Qwen3's task line (<code>asQuery</code> in <code>tutor.ts</code>); saved answers are still embedded as they are, as the model expects. Measured on the demo data: the unrelated answer moved to 0.53 (outside), the related ones moved closer, unrelated topics (bread, weather) stayed at 0.54 and above. The 0.5 cut-off is kept.	Browser: the query now shows only "Verb at the end". Unit test for <code>asQuery</code> . Commit <code>30f4bc5</code> .

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4	Saving the rule made from two answers failed with "Could not save to the database: new row for relation "tags" violates check constraint "tags_name_check1"", and the page then showed "Rule not found".	Owner, first look	The tutor wrote a whole sentence (107 characters) as the merged rule's topic. A rule's topic is stored as a tag, and the database holds a tag name to 60 characters. Nothing in the app checked that: not the tutor's save dialog (it allowed 200), and not the Grammar page's own topic box either. The save is optimistic, so the page had already moved to the new rule before the database refused it.	Fixed once, where every rule save passes: <code>cleanNames</code> in <code>rules.ts</code> cuts a topic to 60 characters, which covers the tutor dialog, Add rule and the rule editor. Also: the tutor is asked for a topic of one to four words, the reply reader caps it, and the dialog's topic box holds at most 60.	Browser: the merged rule saved, its topic stored, and its link to the conversation recorded. Unit tests for the topic cap. Commit <code>30f4bc5</code> .
5	Renaming a conversation did not keep the new name.	Owner, first look	The name box saved only on Enter; clicking away cancelled, by the plan's design (chosen to avoid saving twice). Clicking away is how people usually finish typing, so the new name was dropped. The database write itself worked (tested directly).	Leaving the box now saves, and Enter saves by leaving the box, so the save happens once; Escape cancels (and no longer closes the phone panel around it). Change in <code>TutorSidebar.tsx</code> ; spec updated.	Browser: renamed by clicking away; the database holds the new name. Commit <code>30f4bc5</code> .

Still to do before release

- Set `TUTOR_SIGNING_SECRET` in Vercel as a Sensitive variable before the code reaches `main`.
- Push the database migration to the live project (with the owner's yes; it deletes the old Conversations chats), then one real check on the live data, then merge.
- After the deploy is live, drop the old `conversation_messages` table.